

Isolated MIDI input / 3.3 V UART

M1 | INDEPENDENT MODULE | SEE PIN GUIDE AT RIGHT

PIN GUIDE: IN = Into this board / OUT = out / I/O = both / GND = 0 V

J202 / MIDI IN
1: DIN4 IN 2: DIN5 IN

J101 / PWR IN / UART OUT
1: 5V IN 2: 3V3 IN
3: GND 0V 4: UART OUT

J203 / CASE BOND
1: CASE 2: GND 0V

U201.3: NC -- LEAVE OPEN

U201
onsemi H11L1M

J101
POWER INPUT + UART OUTPUT TO HOST

+5V 1
+3V3 2
GND 3
4

J203
METAL CASE BOND + GROUND

CHASSIS 1
GND 2

DIN panel socket: contact 4 -> J202.1; contact 5 -> J202.2. Pins 1/2/3 and shell unconnected.
Insulate DIN shell. J203.1 bonds the enclosure; R101 returns ESD to local GND.
Optocoupler LED current -> UART low. The output pull-up is 3.3 V, never 5 V.
Functional ground-loop isolation; transient clamps limit common-mode range. Not a safety isolator.

+3V3 +5V CHASSIS GND

Engineering prototype – bench validation required

Ardor

Sheet: /
File: Ardor_MIDI.kicad_sch

Title: Isolated MIDI input / 3.3 V UART

Size: A3 Date: 2026-10-02

KiCad E.D.A. 9.0.2

Rev: M1

Id: 1/1